

ASZADUR RAHMAN

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EDUCATION

Tezpur University <i>B.Tech in Electronics and Communication Engineering — CGPA: 7.17/10</i>	2023 – Present Assam, India
AJMAL SUPER 40 <i>Higher Secondary (Class XII) — 80.4%</i>	2020 – 2022 Assam, India
Don Bosco High School Dergaon <i>Secondary (Class X) — 88.3%</i>	2020 Assam, India

TECHNICAL SKILLS

HDL & Digital Design: Verilog HDL, RTL Design, Digital Logic, FSMs, Counters, Testbench Development
VLSI / EDA: Cadence EDA Tools, SKY130 PDK, Xschem, Ngspice, Magic VLSI, Netgen, Yosys, Icarus Verilog
Programming: C, C++, Python, Embedded C
Embedded / IoT: ESP32, ESP8266, Wi-Fi, Sensors, Web Server, IoT
Tools: Git, GitHub, VS Code, MATLAB, MS Office

PROJECTS

- 6T SRAM Cell Design & Characterization** [Report] — Cadence Virtuoso, GPDK 45nm
- Designed the 6T SRAM cell schematic and layout in Cadence Virtuoso using the GPDK 45nm technology.
 - Generated VTC and butterfly curves to extract Static Noise Margin (SNM), achieving **642.54 mV**.
 - Performed transient analysis to extract **WAT = 29.09 ps** and **RAT = 125.46 ps**.
- CMOS Inverter Design & Analysis using SKY130 PDK** [Report] — Nov 2025 – Dec 2025
- Designed and verified a CMOS inverter through schematic capture, SPICE simulation, layout, DRC/LVS, and RTL-to-layout validation.
 - Performed VTC, propagation-delay, power, noise-margin, sizing, and parasitic analysis using Xschem, Ngspice, Magic VLSI, Netgen and Yosys.
- Full-Custom & Semi-Custom IC Design Flow** [Report] — Cadence EDA Tools — Training Workshop, 2025
- Gained hands-on exposure to the full-custom IC flow using Cadence Virtuoso, including schematic design, layout, DRC/LVS, parasitic extraction and post-layout simulation.
 - Worked through a semi-custom 4-bit adder flow using Verilog, Incisive, Genus and Innovus, covering RTL simulation, synthesis, floorplanning, placement, CTS, routing and GDSII generation.
- RTL Smart Alarm Clock — Verilog HDL — 2026**
- Developed modular RTL for timekeeping, counters, alarm functionality, and control operations using Verilog HDL.
 - Created dedicated testbenches and simulated/debugged the design using Icarus Verilog.
- JAL-SURAKSHA — Smart India Hackathon 2025 — Proposed Solution** [Report] — IoT + AI
- Proposed an IoT and AI-based solution for smart water-quality monitoring and contamination awareness as part of Smart India Hackathon 2025.
- ESP32-Based Wi-Fi IoT Monitoring System — Embedded Systems**
- Developed an ESP32-based IoT system integrating temperature, humidity, light and motion sensors for real-time monitoring using Embedded C/C++.
 - Implemented a Wi-Fi web server to collect and display sensor data through a web interface.

EXPERIENCE / LEADERSHIP

- Convenor — eSports Club, Tezpur University** 2025 – Present
- Led club operations, coordinating tournaments, scheduling, teams, volunteers and event execution.
- eSports Coordinator — TechXtra, Tezpur University** 2024
- Coordinated an eSports event with NODWIN Gaming and KRAFTON, including participant coordination, scheduling and on-ground operations.
- NSS Department Representative — Tezpur University** 2025 – 2026
- Supported departmental NSS activities, student coordination and community-service initiatives.

CERTIFICATIONS & TRAINING

Verilog HDL — Hands On — Maven Silicon (2026) — **Full-Custom & Semi-Custom IC Design Flow — Entuple Technologies (2025)**

ACHIEVEMENTS

All Assam 7th Rank — Mouchaq Scholarship Test; **Scholarship Awardee** — HRD Scholarship Test.
2nd Place — School Science Exhibition Project; **2nd Place** — School Marathon; awards in school athletic events.